Igor Augusto de Melo

Aeronautical Engineering student | Data Analysis
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Summary

Aeronautical Engineering student at EESC-USP (expected 2027) with a **data analysis** internship in **Python and SQL (MySQL on Google Cloud)** and a quantitative engineering background: **CFD and LES** simulations on HPC (Fortran), **Monte Carlo** analysis with 5,000 runs, a **neural network** implemented in Python/NumPy and MATLAB, and **Machine Learning** and **Reinforcement Learning** courses (Data ICMC-USP). Built a personal **AI email triage** system with the Claude API. Exchange year at École Centrale de Marseille; project manager of an autonomous UAV at SEMEAR.

Education

- **Undergraduate degree in Aeronautical Engineering**, *EESC, University of São Paulo* — 2020 – expected 2027
- **Academic exchange, general engineering program**, *École Centrale de Marseille, France* — Aug 2022 – Sep 2023
- **Machine Learning and Reinforcement Learning courses**, *Data ICMC-USP* — 2025
- **Special Studies Certificate in Computational Fluid Dynamics**, *ICMC-USP* — 2023 – present

Experience

Data Analyst Intern — Dec 2024 – Mar 2025
Pilea Labs, influencer and affiliate marketing platform for e-commerce

- Designed and built the **MySQL** databases on **Google Cloud** and the queries behind the platform's charts.
- Joined sales by influencer coupon with social media metrics, at a scale of **thousands of orders**.
- Proposed about 15 dashboard KPIs, delivered **real-time** analyses and integrated **Python** scripts into the website.

Project Manager, autonomous fixed-wing UAV — Apr 2025 – present
SEMEAR Aerial Robotics Group, EESC-USP

- Manage a project in **preliminary design**: a 14-person team across 5 subsystems.
- Define the architecture and **technical requirements** and lead validation using **NASA systems engineering**.
- Run performance and control-oriented **simulations** to support design decisions.

Team Member: propulsion, flight mechanics and systems engineering — Mar 2020 – Dec 2023
TOPUS Projetos Aeroespaciais, EESC-USP

- Built 6-DOF flight simulation tools in **MATLAB** and ran **5,000 Monte Carlo simulations** (thrust, drag, wind) to map the landing area.
- Compared simulated and measured results: simulated apogee of 1,160 m against **1,168 m reached**.
- Applied **NASA** systems engineering methods to project management and documentation standards.

Data, AI and Automation Projects

Email triage — Python, Claude API
Personal project, 2025 – present

- **Claude Haiku** through the Claude API classifies emails into JSON (folder, summary, suggested action, deadline), with prompt caching.
- Python pipeline with OAuth2, IMAP, cron and Telegram digests, running on a VPS. Estimate: 1 to 3 min saved per email.

Neural network — Python/NumPy, MATLAB
Undergraduate research, EESC-USP

- Implemented, without ML libraries, a multilayer **neural network** with configurable layers and neurons, aiming to optimize the parameters of the Beddoes-Leishman and ONERA semi-empirical models of unsteady aerodynamic loads, calibrated against experimental and **CFD** data.

Trilingual portfolio — GitHub Pages, Claude Code
Personal project

- PT/EN/FR site built with **Claude Code**, with language-parity, link and résumé-sync tests.

Research

Turbulent flow simulation Sep 2023 – Dec 2024

ICMC-USP, FAPESP grant

- Simulated turbulent channel flow with Coriolis force using **LES** (Re_τ = 180, 64 × 64 × 64 grid), compared with **DNS**.
- Developed and ran the numerical routines in **Fortran** on HPC; results published at EPTT 2024.

Atmospheric modeling and wind energy research Feb – Jul 2023

IRPHE, École Centrale de Marseille, France

- Set up and ran the **WRF** weather model on three nested domains (Python, Bash, Linux).
- Analyzed wind-related **uncertainty** in a wind turbine's energy output (defense slides, in French).

Publication

Igor Augusto de Melo, Livia S. Freire, Leandro Franco de Souza, Matheus Tozo de Araujo. *Turbulent Channel flow with Coriolis force using Large-Eddy Simulation*. 14th Spring School on Transition and Turbulence (EPTT 2024), São João da Boa Vista, SP, Brazil, Sep 2024. Paper (PDF).

Skills

- **Data:** Python (NumPy), SQL (MySQL), data analysis, KPIs and dashboards, Monte Carlo simulation, Machine Learning and Reinforcement Learning (coursework), neural networks.
- **AI and automation:** Claude API and prompt design, Python automation, OAuth2, IMAP, cron.
- **Programming and environment:** MATLAB/Simulink, Fortran, Bash, C++ (basic), Java (basic); Git, Linux, HPC, Google Cloud, LaTeX.
- **Engineering:** CFD (ANSYS Fluent/CFX, OpenFOAM), aerodynamics, flight mechanics, systems engineering (NASA SE Handbook).

Languages and Other Activities

- **Languages:** Portuguese (native), English (advanced), French (advanced), German (basic), Spanish (basic).
- **Volunteering:** Gatos do C2, USP São Carlos (Mar – Oct 2024); coordinated rescue logistics for the spay/neuter, care and release program.